Grade 10 · Algebra II · Topic 2: Quadratic Functions and Equations · Lesson 2-4 — Overview of Complex Numbers

# Project Task — NEOM Pump Circuit

*A 20-minute group project. Work in fours. Every member signs the sheet.*

| | |
|---|---|
| Student name : ..... | Date : ..... |
| Group members : ..... ..... ..... ..... | |

**MATERIALS** This sheet · Squared paper · A pencil and ruler · A calculator

## THE DRIVING QUESTION

**Two pump circuits on a NEOM desalination site are connected to the same 120 V supply. Which circuit draws the bigger current — and is a 30 A breaker enough for both?**

## THE SITUATION

A classroom model (invented for practice, not plant data): each pump has two parts in series, so the impedances add. Pump A has $2 + 3i$ and $1 + i$ ohms; Pump B has $4 + 2i$ and $1 + 3i$ ohms.

The current is $I = \dfrac{V}{Z}$ with $V = 120$ V. Engineers need the current in the form $a + bi$ and its size (modulus) in amperes.

Each group then decides whether one 30 A breaker is enough for each pump.

$$V = 120 \text{ V} \qquad I = \frac{V}{Z_1 + Z_2}$$

| Pump | Z (ohms) | I = a + bi (A) | \|I\| (A) |
|---|---|---|---|
| A | | | |
| B | | | |

## YOUR TASK 5 steps · 20 minutes

**1 — ADD THE IMPEDANCES (4 min)**
Find $Z$ for Pump A and Pump B by adding real parts and imaginary parts separately.

**2 — FIND THE CURRENT (5 min)**
Write $\dfrac{120}{Z}$, multiply top and bottom by the conjugate of $Z$, and give $I$ in the form $a + bi$.

**3 — FIND THE SIZE (4 min)**
Find $|I|$ for each pump and check it as $\dfrac{120}{|Z|}$.

**4 — DECIDE (4 min)**
Compare each $|I|$ with the 30 A breaker. Which pump has the smaller margin, and by how many amperes?

**5 — PRESENT (3 min)**
One sentence for the site engineer: why does the conjugate make the division possible?

## YOUR WORKING write on this sheet

### Step 1 — the total impedances:

### Step 2 — the currents in the form a + bi:

### Step 3 — the sizes and the check:

### Steps 4 and 5 — the breaker decision and our sentence:

**ROLES**

- Reader — states the model and the supply
- Calculator — adds the impedances and finds the currents
- Checker — verifies each current by multiplying back
- Presenter — says the sentence

**DONE WHEN ...**

- The conjugate is written before the division is simplified.
- Each current is checked: current times impedance returns $120$.
- Units (ohms, amperes) appear in every answer.
- Your sentence makes sense to someone who does not study maths.